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The Real Cost of Power: Commercial and Industrial Energy Burden Across the World

Kilambi Research

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Abstract. Industrial electricity prices across major economies span a 15-fold range, from $0.02/kWh in Qatar to $0.29/kWh in Ireland. This paper argues that the nominal tariff is a poor measure of the burden firms actually bear. Assembling all-in delivered prices (energy, demand charges, taxes, carbon), reliability-adjusted effective costs, subsidy positions, and sector energy intensities for 54 economies, it builds a composite burden index and income-adjusted burden metrics and tests a central hypothesis: low-nominal-price economies can face greater effective burden, while high-price economies can partially offset cost through productivity and efficiency. The hypothesis survives but with sharp limits. Rankings by nominal price carry almost no information about real burden: the rank correlation between nominal industrial price and income-adjusted burden is 0.30, and against the composite burden index it is 0.01. Nigeria's $0.049/kWh commercial tariff coexists with an effective burden 5 to 10 times higher once self-generation is counted; Ireland's $0.29/kWh industrial price, the sample's highest, maps to the second-lightest burden once low energy intensity and near-zero outages are counted. Aluminium smelting inverts the ranking completely (Qatar $280/t vs Germany $3,745/t of energy cost per tonne), while semiconductors show the opposite pattern: enormous physical intensity, negligible cost burden. The paper documents where national tariffs mislead (bilateral contracting, captive power, coal-route chemicals), where volatility rather than price level binds investment (PJM capacity, Korea's frozen fuel adjustment), and where the forward scenarios are most fragile.

Keywords: energy prices, industrial competitiveness, energy burden, electricity tariffs, reliability, subsidies

JEL Classification: Q41, Q48, L94

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1 Introduction

Energy cost comparisons across countries usually begin and end with the tariff: cents per kilowatt-hour, ranked highest to lowest. The ranking is easy to produce and easy to misread. A firm deciding where to smelt aluminium, run a data centre, or build a factory does not pay the tariff. It pays the delivered bill: energy charges plus demand and capacity charges, non-recoverable taxes, carbon costs, and the fuel surcharges that regulators add between tariff reviews. It pays for the power that does not arrive, through backup generation, spoiled batches, and idle shifts. It pays the financing cost of waiting for a connection. And in many economies it pays a policy price, set below or above economic cost by subsidy, cross-subsidy, or frozen adjustments that unwind later, sometimes abruptly.

This paper measures the commercial and industrial energy burden across 54 economies with that fuller definition. The dataset covers all-in industrial and commercial electricity prices, delivered gas prices, wholesale power prices, demand charges, carbon costs, subsidy flags, and reliability indicators, with price histories for 2016, 2021, and 2026 in local currency, US dollars, and real terms. Sector modules translate tariffs into costs per tonne of steel, aluminium, chemicals, cement, paper, per wafer, per megawatt of data-centre load, and per square metre of commercial floor space. A volatility scorecard records how customers actually experience price risk, and a scenario module sets out base, high-cost, and low-cost paths for the next five to ten years.

The organising hypothesis, stated before estimation and tested against the data, is that the effective burden diverges systematically from the nominal price: low-nominal-price economies can face greater effective burden (lower value per unit of energy consumed once reliability, intensity, and subsidy are counted), while high-price economies can partially offset cost through productivity and efficiency. Section 9 reports where the hypothesis holds, where it fails, and what would have falsified it.

Three related Kilambi Research papers anchor parts of the analysis. The time-to-power paper (Kilambi Research, 2026a) supplies the financing-cost framework for connection delays, verified rather than carried over. The AI datacenter buildout paper (Kilambi Research, 2026b) supplies load scenarios for the forward assessment, likewise re-verified. The Malaysia market study provides demand-charge anchors, checked against the Energy Commission schedule before quoting.

The paper is structured as follows. Section 2 defines the burden and the five wedges between tariff and burden. Section 3 describes the data. Section 4 maps the global price landscape. Section 5 tests nominal against effective. Sections 6 and 7 work through sectors and reliability. Section 8 covers volatility and predictability. Section 9 sets out forward scenarios. Section 10 states the falsifiers and limitations. Section 11 concludes.

2 What "burden" means

2.1 The five wedges

The burden $B_{c,s}$ for country c and customer segment s is the all-in cost of usable energy services per unit of economic output, not the posted price per kilowatt-hour. Five wedges separate the two:

1. **Delivered vs posted.** Demand and capacity charges, wires charges, non-recoverable taxes and levies, and carbon costs sit on top of the energy charge. In Mexico's GDMTH industrial tariff, demand charges near 760 MXN/kW-month are roughly 51% of the pre-tax bill at 60%

load factor; quoting the 1.10–1.97 MXN/kWh energy charge alone understates the delivered cost by about half (CALCULATED, B). Commercial bills in the United States are 30–70% demand charges (McLaren et al., 2017).

2. **Reliability.** Unserved energy is replaced at backup cost (typically $0.30–0.60/kWh for diesel self-generation) or lost as output. The effective price is a weighted average of grid and backup cost, and the weights are outage hours.

3. **Subsidy and cross-subsidy.** Administered prices are fiscal positions, not cost advantages. Ecuador's $0.10/kWh regulated tariff sits against $0.24–0.25/kWh thermal barge cost and $0.88/kWh import cost in September 2026 (OBSERVED, C); Pakistan's tariff differential subsidy feeds circular debt near Rs2.5tn (OBSERVED, B). The burden surfaces later as taxes, arrears, or curtailment.

4. **Intensity and productivity.** The same kilowatt-hour buys more output where processes are efficient and value added per unit of energy is high. Ireland's manufacturing pays the sample's highest industrial price and among its lowest energy costs per $1M of manufacturing value added.

5. **Volatility and predictability.** A stable $0.15/kWh and a $0.15/kWh average with 40% annual swings are different investment propositions. Section 8 documents where the path, not the level, binds.

2.2 The composite burden index

To rank economies on effective rather than nominal burden, the paper computes a composite index for 47 of the 54 countries (seven lack an industrial all-in price and are excluded; Section 3). The index is the mean of three min-max normalised log components, scaled 0–100:

$$I_c = 100 \times \frac{1}{3} \left[\tilde{p}_c + \tilde{e}_c + \tilde{o}_c \right], \tag{1}$$

where $\tilde{p}_c$ is normalised log industrial all-in electricity price, $\tilde{e}_c$ is normalised log electricity intensity (economy-wide electricity consumption in kWh per US dollar of manufacturing value added), and $\tilde{o}_c$ is normalised log of one plus annual outage hours. Higher is heavier. The index is (CALCULATED, C): a transparent, arbitrary-weighted summary, not a structural estimate. Its components are published alongside it so any reader can reweight. Where outage hours are missing, the index averages the two available components; those cases are flagged.

Two cautions apply before any ranking is read. First, the intensity component uses economy-wide electricity consumption, not industrial consumption, because consistent industrial-consumption series do not exist for all 54 countries; it is a proxy for how much electricity the economy uses per unit of manufacturing value, not a budget share. Values above 100% of manufacturing value added (Ukraine, Australia) confirm it cannot be read as a share. Second, equal weights are a choice, and Section 10 shows the ranking's sensitive cases.

3 Data and coverage

The program dataset comprises 54 countries across six continents: 108 snapshot rows (industrial and commercial segments), 247 history rows (2016, 2021, 2026 in local currency, USD, and 2026 real

local terms), macro context for all 54, a 54-country reliability file, sector cost sheets, a 41-market volatility scorecard, and scenario notes. Snapshot vintage is 2025 for 42 countries and 2026 for 12; the year column in every file carries the vintage, and the paper does not present mixed vintages as same-year without the caveat.

Sources are producer-first throughout: IEA, World Bank, IMF, OECD, Eurostat, US EIA, UK DESNZ, national regulators, utility tariff schedules, the World Steel Association, the International Aluminium Institute, SEMI, company filings, Uptime Institute, CBRE, and the ElkassabgiData Econ Data Library as one source among several for macro, value-added, and price series (International Energy Agency, 2025a; World Bank, 2024; U.S. Energy Information Administration, 2025; Eurostat, 2025; International Monetary Fund, 2025; Department for Energy Security and Net Zero, 2025; World Steel Association, 2025; International Aluminium Institute, 2025; SEMI, 2025; CBRE Research, 2025; Uptime Institute, 2024). Every row carries a source citation, a data-through date, and an evidence grade on the unified vocabulary: Observed, Calculated, or Modelled, graded A, B, or C. Grades travel into the paper's tables and figure notes.

Known gaps are listed, not filled: no industrial coal or coke prices (Section 6.1); demand-charge bill shares absent from snapshot tariffs; five countries with no SAIDI data at all; industrial all-in prices missing for France, Italy, Spain, the Netherlands, Poland, Norway, and Egypt; DESNZ Table 5.4.2 unrecoverable after the UK departmental reorganisation; and the Malaysia demand-charge capacity/network split still to confirm against the Energy Commission schedule. US and Canadian 2024 manufacturing value added are estimated by holding 2021 WDI shares constant (Calculated, C).

4 The global price landscape

Figure 1 ranks the 47 countries with an observed industrial all-in electricity price. The range runs 15-fold, from Qatar ($0.0192/kWh) and Kuwait ($0.0163/kWh) to Ireland ($0.2884/kWh), Australia ($0.266/kWh), and Singapore ($0.269/kWh) (Observed, B/C). The familiar developed-economy cluster (Germany $0.2558, United Kingdom $0.2275, Philippines $0.215, Colombia $0.231) sits at the top; the Gulf, Russia ($0.0488), and parts of Southeast Asia at the bottom. The United States illustrates why national averages mislead: the 2025 industrial average of 8.61c/kWh spans Texas at 6.55c to California at 21.36c, a 3.3x intranational spread (Observed, B).

Three structural facts sit behind the ranking. First, wholesale prices are a minority of the delivered bill in most markets once capacity, transmission, taxes, and levies are added: PJM's all-in wholesale averaged $79.28/MWh over January–September 2025 against a $49.28/MWh headline energy LMP (Observed, B). Second, carbon pricing is now a material wedge in Europe (EU ETS averaged EUR 73.43/t in 2025) and a negligible one in most of Asia and the Americas. Third, several low headline prices are policy prices: Gulf tariffs are administered fuel prices, and Bangladesh's BPDB sells near Tk4.5/unit below cost, a fiscal transfer recorded as cheap power.

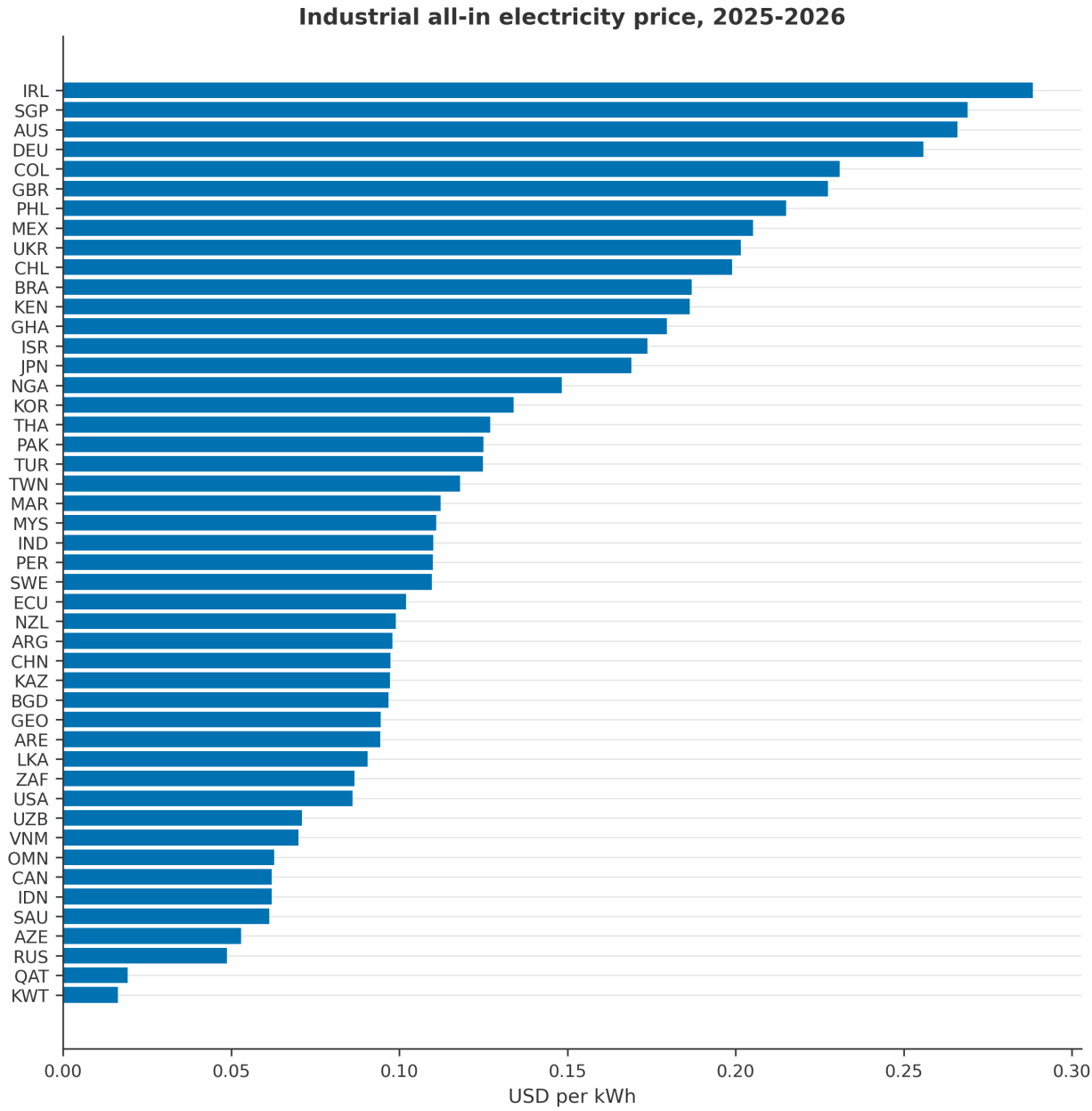


Figure 1. Industrial all-in electricity prices, 47 countries, 2025–2026 vintages. Bars start at zero. Countries without an observed industrial all-in price are excluded.

5 Nominal vs effective: where the tariff misleads

5.1 The burden index ranking

Figure 2 ranks the 47 countries on the composite burden index. The top of the ranking, heaviest burden, reads: Ukraine (88.9), Nigeria (75.0), Georgia (73.6), South Africa (72.5), Pakistan (69.8), Kazakhstan (68.9), Kenya (65.7), Chile (65.7). The bottom, lightest burden: Qatar (34.3), Ireland (39.4), Taiwan (39.4), South Korea (42.6), Kuwait (42.8), Germany (43.9), Singapore (45.0) (Calculated, C).

Figure 3 plots the index against the nominal price. The correlation is positive but loose, and the off-diagonal cases are the paper's subject. Ireland has the sample's highest nominal industrial price and its second-lightest burden: low electricity intensity per dollar of manufacturing value added and 3.65 outage hours a year absorb the tariff. Germany ($0.2558/kWh, index 43.9) and Singapore ($0.269/kWh, index 45.0) tell the same story. In the other corner, Nigeria's $0.1483/kWh industrial tariff (and $0.049/kWh commercial tariff, the sample's cheapest) maps to the third-heaviest burden, driven by an estimated 4,500 unserved hours a year and the backup economics of Section 7. South Africa ($0.0866/kWh, index 72.5) and Pakistan ($0.125/kWh, index 69.8) are heavy-burden, mid-price economies.

The falsifier for this exercise was stated in advance: had the index correlated near-trivially with the nominal price, the "effective burden" framing would be relabelling. It does not, and the divergences are economically interpretable. But the index has sensitive cases, and Section 10 names them: Ukraine's war-distorted intensity, Nigeria's estimated outage hours, and the equal weights.

5.2 Income-adjusted burden: the ranking inversion, stated systematically

The misleading-nominal cases above are instances of a systematic pattern. Figure 11 ranks the 47 countries by the cost of 1 MWh at the industrial all-in tariff as a share of GDP per capita (nominal), the income-adjusted burden (Calculated, B). The spread is far wider than the nominal one: about 18x in nominal price (Qatar $0.0192 to Ireland $0.2884/kWh) becomes nearly 400x income-adjusted, from 0.03% of GDP per capita in Qatar to 12.0% in Kenya. On a PPP basis the range is 0.02% (Qatar) to 4.8% (Kenya). The heaviest income-adjusted burdens fall on Kenya (12.0%), Ghana (9.4%), Pakistan (8.4%), Nigeria (7.2%), and the Philippines (5.4%); the lightest on Qatar (0.03%), Kuwait (0.06%), the United States (0.10%), Canada (0.11%), and Sweden (0.17%).

Figure 12 plots each country's rank by nominal price against its rank by income-adjusted burden. If tariffs measured burden, every country would sit on the dashed diagonal. The rank correlation (Spearman) between nominal industrial price and income-adjusted burden is 0.30, weak; between nominal price and the composite burden index of Section 2 it is 0.01, essentially zero. Nominal price rankings carry almost no information about who bears the heaviest real energy burden. The off-diagonal mass has a clear structure: high-income economies with expensive power (Ireland, Singapore, Germany) sit bottom-right, expensive nominally and light in burden; low-income economies with mid-range tariffs (Kenya, Ghana, Pakistan, Nigeria) sit top-left, cheap or mid-priced nominally and heavy in burden. The Gulf sits bottom-left on both: cheap power and light burden, the one quadrant where the nominal ranking does not mislead.

Three caveats bound this result. First, GDP per capita is an income proxy, not a firm's cost base; the preferred denominator, median industrial wages, has no consistent 54-country series in the

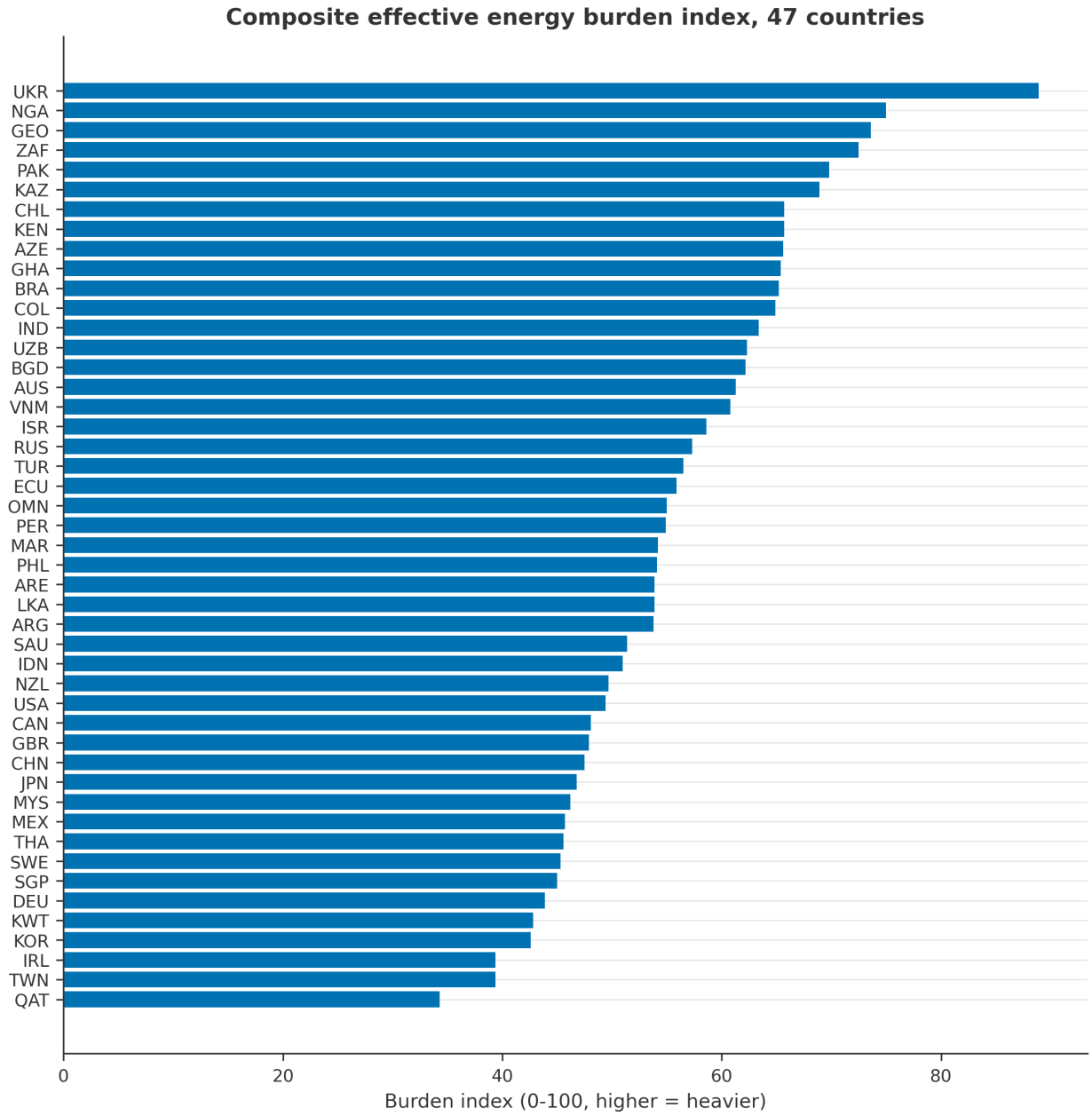


Figure 2. Composite effective energy burden index, 47 countries. Mean of normalised log industrial price, log electricity intensity, and log outage hours, scaled 0–100. Seven countries excluded for missing industrial all-in price.

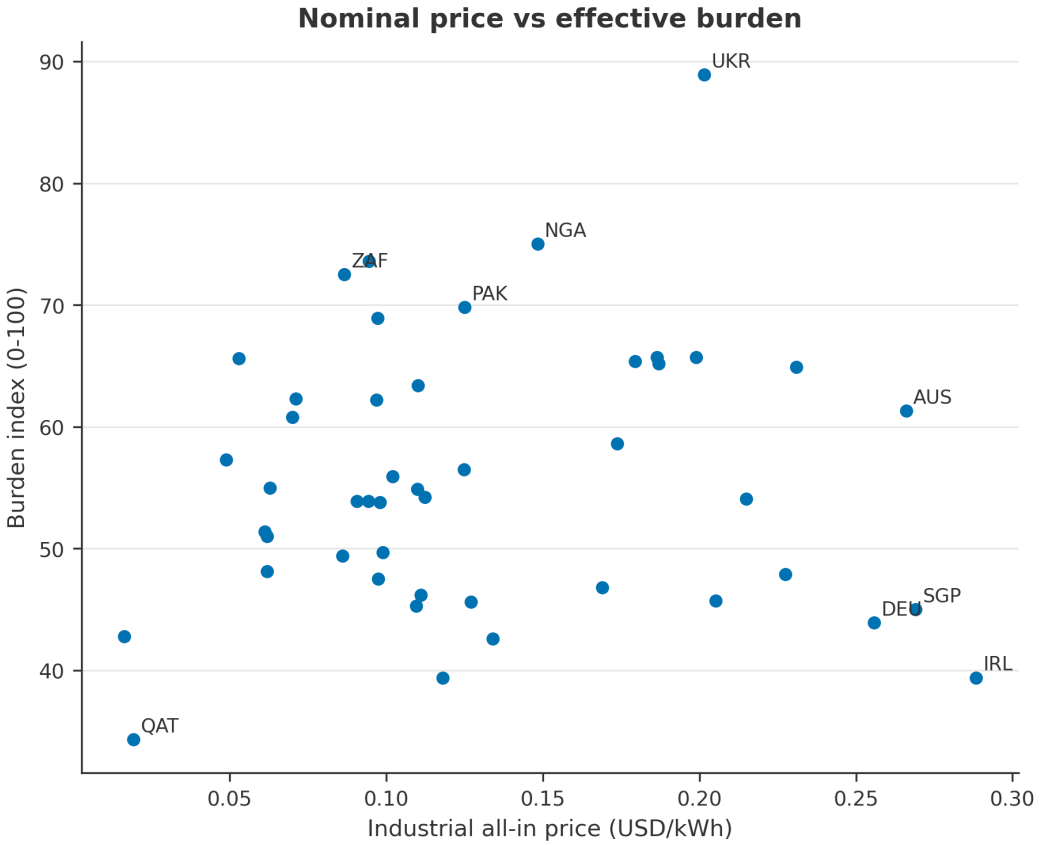


Figure 3. Nominal industrial price vs composite burden index. Off-diagonal economies (top-left, bottom-right) are where the tariff misleads.

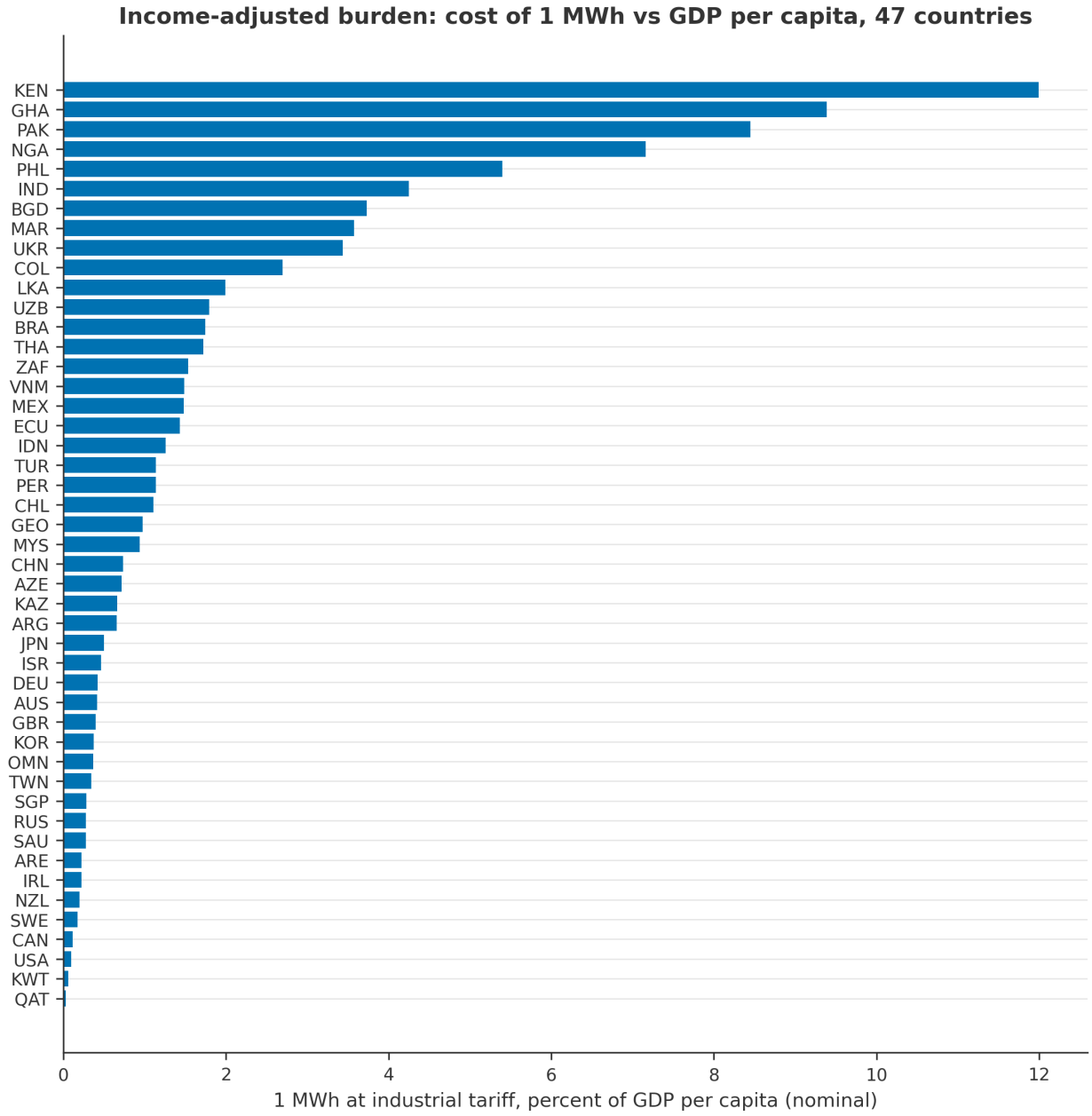


Figure 4. Income-adjusted burden: cost of 1 MWh at the industrial all-in tariff as a share of GDP per capita (nominal), 47 countries. Bars start at zero.

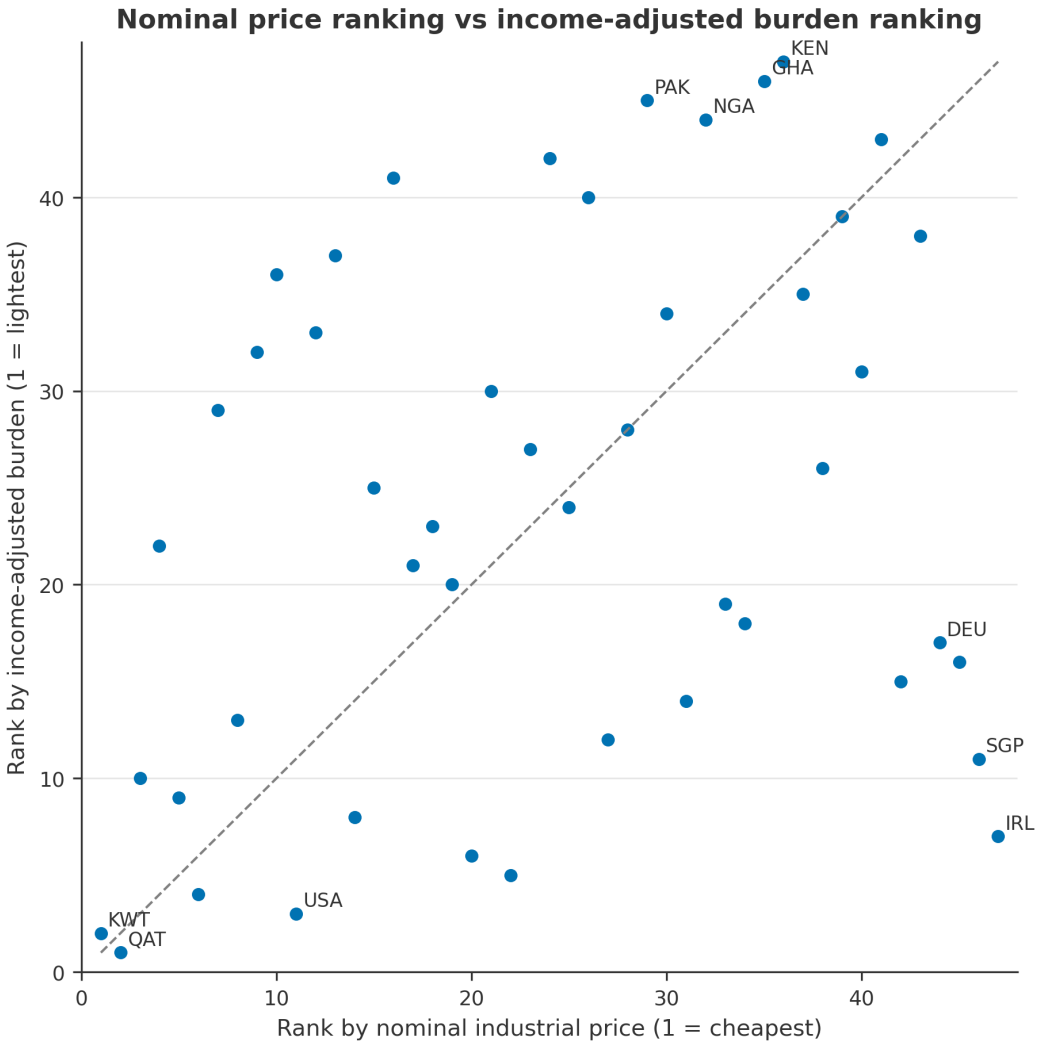


Figure 5. Nominal price ranking vs income-adjusted burden ranking, 47 countries. Distance from the dashed diagonal is the size of the ranking error a tariff table makes.

underlying data sources (ElkassabgiData carries no wage series) and is listed as a data gap for a future refresh. Second, the PPP adjustment compresses but does not remove the pattern: the rank correlation on PPP terms is 0.34. Third, the composite index's near-zero correlation partly reflects its construction: the intensity component (electricity per dollar of manufacturing value added) falls with income, offsetting the price component. That is not a defect of the index; it is the productivity-offset half of the thesis, measured.

5.3 Ten documented misleading-nominal cases

The regional notes files flag ten cases where the headline number most misleads, each with the mechanism:

1. **Nigeria.** The $0.049/kWh commercial tariff is the cheapest in the dataset and the least informative. No public national SAIDI exists; 12 grid collapses occurred in 2024 and 82% of firms report outages. Firms self-generating at $0.30–0.60/kWh face an effective burden 5 to 10 times the tariff (Calculated, C). This is the cleanest test of the paper's thesis, and the thesis passes it.

2. **South Africa.** At the 2023 load-shedding peak (1,296 hours), the effective commercial cost was $0.186/kWh against a $0.140 tariff (+33%); at the improved FY2025 level (175 hours), +4%. The tariff barely moved; reliability moved the burden by a third (Calculated, B).

3. **Sweden.** The cheapest industrial electricity in the European sample (EUR 0.0970/kWh) alongside the EU's highest industrial gas price (EUR 0.1065/kWh). Any dual-fuel burden calculation using electricity alone is wrong for Sweden (Observed, B).

4. **Mexico.** GDMTH energy charges of 1.10–1.97 MXN/kWh look cheap; demand charges near 760 MXN/kW-month are about 51% of the pre-tax bill at 60% load factor, making the $0.2052/kWh all-in the highest industrial rate in the North America set (Calculated, B).

5. **Ecuador.** The $0.10/kWh regulated tariff sits against $0.24–0.25/kWh thermal barge cost and $0.88/kWh import cost in September 2026; the burden surfaces as curtailment of roughly 150 large consumers rather than as price (Observed, C).

6. **China.** The 0.40 CNY/kWh coal benchmark hides a 2–3x provincial spread (Yunnan ∼0.35 vs Guangdong ∼0.75 CNY/kWh) (Observed, B). The national average is not a site price.

7. **United Kingdom.** The 17.3p/kWh manufacturing average vs the 26.6p/kWh IEA medium-consumer figure is segment definition (large users, climate-change agreements, self-generation), not a data error (Observed, B).

8. **India.** The IEX day-ahead price ($51/MWh) is about 7% of the market and roughly half the delivered industrial price (Observed, B).

9. **Pakistan.** The $0.125/kWh average industrial tariff overstates the marginal price: expansion load under Roshan Maeeshat pays about $0.08/kWh (Observed, B). Average and marginal diverge.

10. **Poland.** PLN-denominated wholesale looks moderate against German levels, but a 36% non-recoverable tax share (the EU's highest) plus full ETS pass-through on a coal-heavy system makes the delivered burden heavier than the wholesale price suggests (Observed, B).

Figure 8 shows the dual-fuel dimension directly: industrial gas against industrial electricity prices. Sweden is the visible outlier, top-left in gas terms. The Gulf sits bottom-left on both. Economies in the top-right (Germany, the United Kingdom, the Philippines) face the heaviest combined thermal-and-power burden.

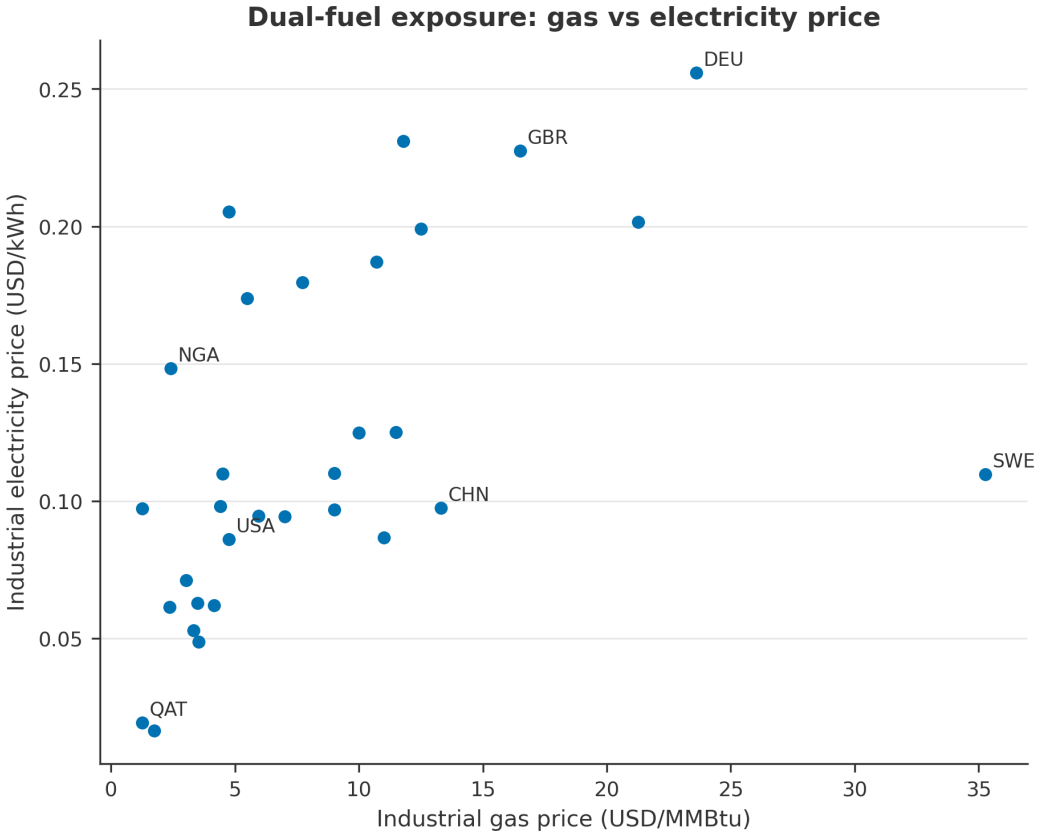


Figure 6. Industrial gas vs industrial electricity prices. Sweden combines the sample's cheapest industrial electricity with its most expensive industrial gas.

6 Sector deep dives: the tariff is not the cost

National tariffs are averages over customer classes, contracts, and fuels. Sectors translate them into costs per unit of output, and the translation is where competitiveness is decided. Four sector modules follow.

6.1 Energy-intensive industries: aluminium inverts the ranking

Figure 4 shows energy cost per tonne of primary aluminium across 11 countries, computed at typical smelting intensity (∼14.5 MWh/t) times the national industrial tariff (CALCULATED, B). Because smelting intensity is technology-fixed, the tariff passes through one for one, and the ranking inverts the usual development order: Qatar $280/t (15% of a reference cash cost), Russia $713/t (38%), the United States $1,256/t, India $1,610/t (85%), Germany $3,745/t (197%), Australia $3,857/t (203%).

German energy cost is roughly five to six times the sector's value added per tonne. The European curtailments of recent years (Slovalco, Aldel, San Ciprian) are exactly what these numbers predict; for aluminium, the nominal price ratio is the whole story.

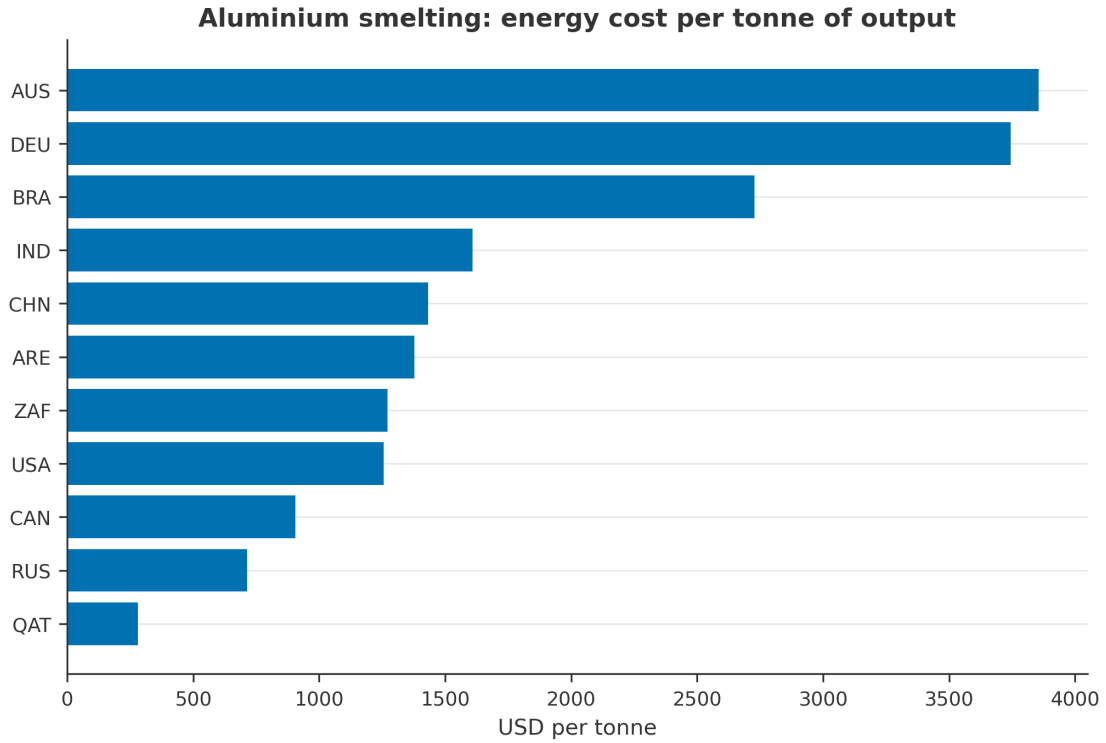


Figure 7. Primary aluminium: energy cost per tonne at national industrial tariffs, typical intensity 14.5 MWh/t. Bilateral and captive contracts (see text) make several national-tariff rows upper bounds.

But the chart also demonstrates the paper's sharpest internal critique: for aluminium specifically, the national tariff is often the wrong number. Indian smelters run on captive coal power at roughly \$40–50/MWh against an 11c/kWh grid tariff; South Africa's Hillside smelter holds a negotiated deal near \$0.034/kWh against 8.7c national; Gulf smelters run on dedicated gas-fired power below headline tariffs; surviving US smelters live on legacy contracts while greenfield entry needs sub-\$40/MWh deals the tariff does not provide (Observed, C). The country ranking is valid for the tariff; the plant ranking can differ by a factor of two or more. The paper reports both and does not let the chart stand alone.

Steel, cement, pulp and paper, and chlor-alkali show milder versions of the same pattern. Two measurement traps recur. First, the coal blind spot: ammonia at German gas prices costs \$785/t of energy (245% of reference cash cost, matching the 2022–23 European shutdowns) against Qatar at \$43/t, but China's \$428/t row is notional because roughly 75% of Chinese ammonia is coal-gasification based; pricing it at China's \$13.3/MMBtu gas price hugely overstates actual Chinese cost (Calculated, C). The same applies to urea and cement kilns, which burn coal, not the gas the dataset prices. Industrial coal and coke prices are the dataset's largest sector gap, listed as such in Section 10. Figure 10 shows the gas-route chemicals rows with the notional warning in the title. Second, the self-generation blind spot: Brazilian kraft mills are energy self-sufficient via black liquor and often net exporters, so purchased-power math (\$93.5/t at the 18.7c tariff) is an upper bound, not a level (Observed, B).

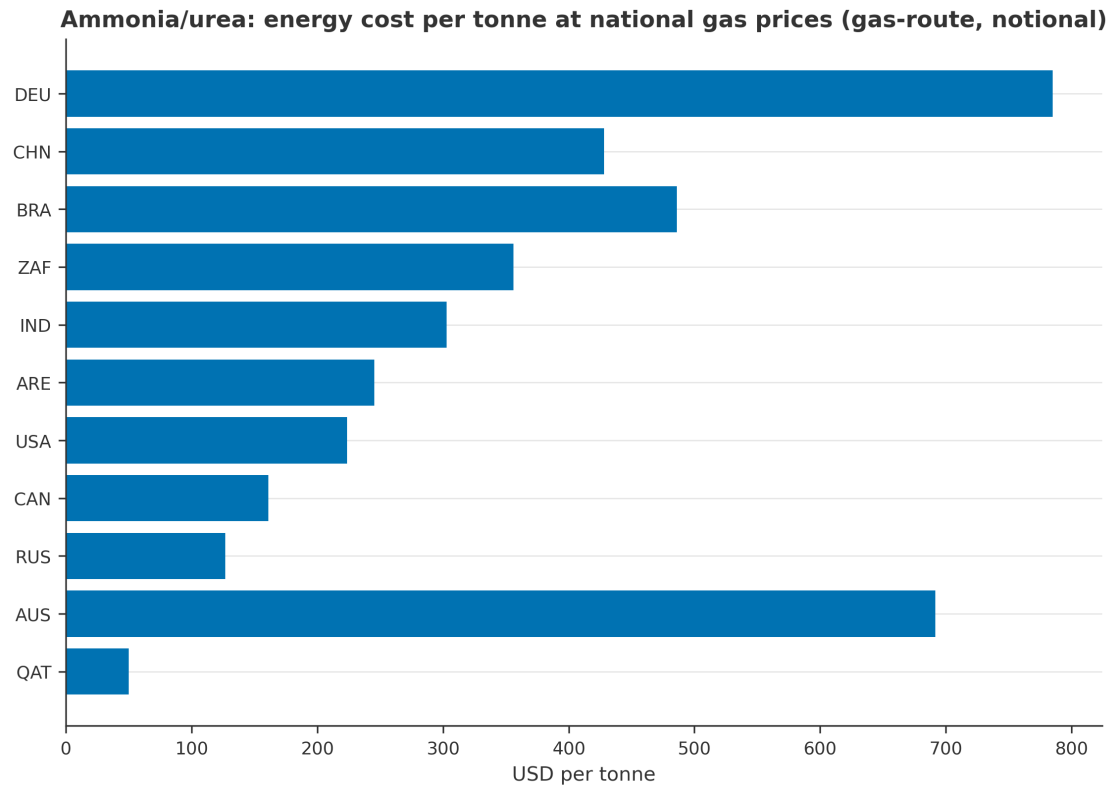


Figure 8. Ammonia and urea: energy cost per tonne at national industrial gas prices. Gas-route only; coal-based production (notably China) is not represented. Rows are notional where gas is a thermal proxy.

6.2 Data centres and semiconductors: tariffs vs physics

Fleet-average power usage effectiveness has been stuck at 1.56 for five years (Uptime Institute, 2024), so cross-country data-centre economics are almost entirely tariff-driven. Figure 5 shows the annual grid electricity cost per megawatt of IT load at PUE 1.56: Ireland ∼$3.9M, Singapore ∼$3.7M, Germany ∼$3.5M, Australia ∼$3.6M, against China ∼$1.3M, the United States ∼$1.18M, and Texas ∼$0.87M (Calculated, B). A new-build PUE-1.3 facility in Singapore (∼$3.06M/yr) still costs 3.1 times a PUE-1.56 US facility: efficiency cannot close the tariff gap. The policy overlay matters as much as the price: Ireland's large-energy-user connection policy forces 100% on-site flexible generation and 80% new renewables, making connection rules rather than price the investment constraint; Malaysia passes RM43bn of RP4 grid capex through the tariff base, with demand charges at 26–32% of the bill, among the sample's highest (Observed, B/C).

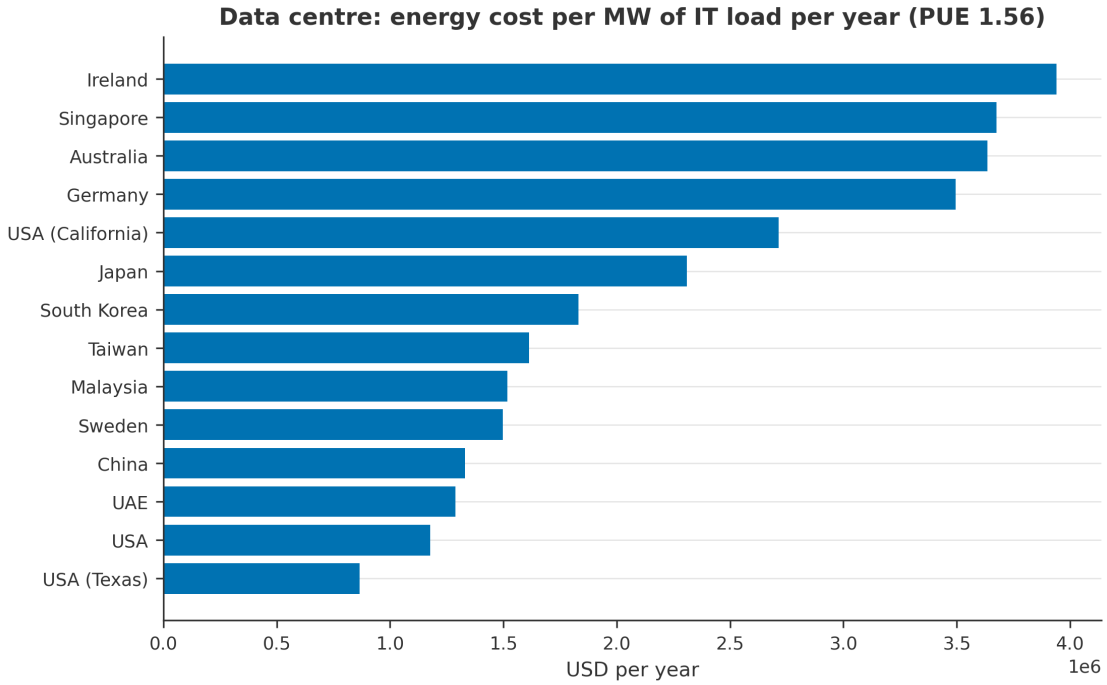


Figure 9. Data centres: annual grid electricity cost per MW of IT load at fleet-average PUE 1.56. Efficiency (PUE) varies little across countries; the tariff is the business case.

Semiconductors show the opposite pattern: enormous physical intensity, negligible cost burden. TSMC's ∼25.5 TWh of purchased electricity is 8–9% of Taiwan's national consumption (Taiwan Semiconductor Manufacturing Company, 2025), and per-wafer energy is rising with node shrinks (per-mask-layer energy rose from 27.7 kWh in 2022 to 40.5 kWh in 2023 with the 3nm ramp, implying ∼4 MWh/wafer at leading edge against under 1 MWh for mature nodes), yet the ∼$3B energy bill is under 3% of revenue (Observed, B). Tariff differentials (Germany ∼$506/wafer vs the United States ∼$170) matter only at the margin for new-fab siting. The burden story for chips is node-shrink escalation, not geography.

6.3 Mining and food: geology and heat

In mining, geology is a tax the tariff cannot offset. Chile's copper head grade fell from 0.93% to 0.62% between 2005 and 2024, pushing electricity per tonne of copper from 3.07 to 5.16 MWh (+68%); at Chilean intensity, the Australia–China tariff spread turns ∼$503/t Cu (China tariff) into ∼$1,373/t Cu (Australia tariff), a 2.7x multiplier on a commodity where energy is a large cost share (Calculated, C). Nickel laterite via RKEF needs 7,000–8,000 kWh/t Ni, over six times sulphide-route intensity.

Food processing shows a 10x spread inside one sector driven by heat, not electricity: milk powder at 2,801 kWh/t against fluid milk at 281 kWh/t, with drying and evaporation (thermal energy) as the driver. Indicative energy cost per tonne of milk powder runs ∼$241 (United States) to ∼$745 (Australia), but these are electricity-tariff proxies graded (Calculated, C): gas price is the true driver, and electricity-tariff comparisons mislead for dairy. The dairy benchmarks are 2004-vintage, a stated limitation.

6.4 Commercial buildings

Figure 9 translates commercial tariffs into annual energy cost per square metre for hospitals and offices across 10 countries (Calculated, B). A hospital runs ∼$15/m2/yr in Nigeria against ∼$76/m2/yr in Kenya; an office ∼$11/m2/yr in Saudi Arabia against ∼$43/m2/yr in Singapore. Typical electricity intensities: warehouses 60–250 kWh/m2/yr (typical 110), offices 90–350 (170), hotels 100–350 (150), hospitals 180–600 (300), shopping centres 120–350 (200), airport terminals 140–320 (230), supermarkets 350–600 (450), all with climate-zone caveats. Two warnings: commercial bills are 30–70% demand charges (McLaren et al., 2017), which snapshot tariffs do not capture, so per-m2 costs are load-factor sensitive; and energy-cost shares of opex are not available for hospitals, malls, or airports and are not fabricated.

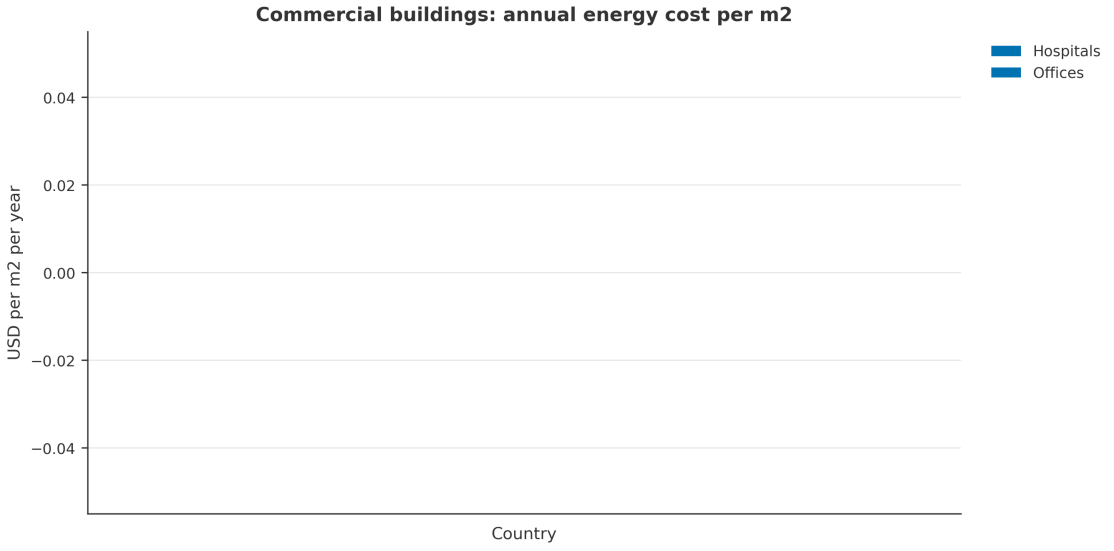


Figure 10. Commercial buildings: annual energy cost per m2 at commercial all-in tariffs, typical intensities. Load-factor sensitivity (demand charges) not captured.

7 Reliability as cost

Figure 7 compares grid tariffs with effective all-in costs including backup generation at an illustrative $0.45/kWh for five countries with observed outage data (Calculated, C). South Africa's 2023 peak load-shedding year adds a third to the tariff; Pakistan's 867 hours add about 21%; India's 380 hours add about 10%. Kenya and the Philippines show the inverse: high tariffs ($0.253 and $0.215/kWh) with low measured outage hours, so backup barely changes the number and the tariff is the burden. Nigeria, the most important case, cannot be plotted: no public national SAIDI exists, and the 5–10x effective-burden estimate rests on firm-survey evidence and the 12 grid collapses of 2024 rather than a single outage-hours series.

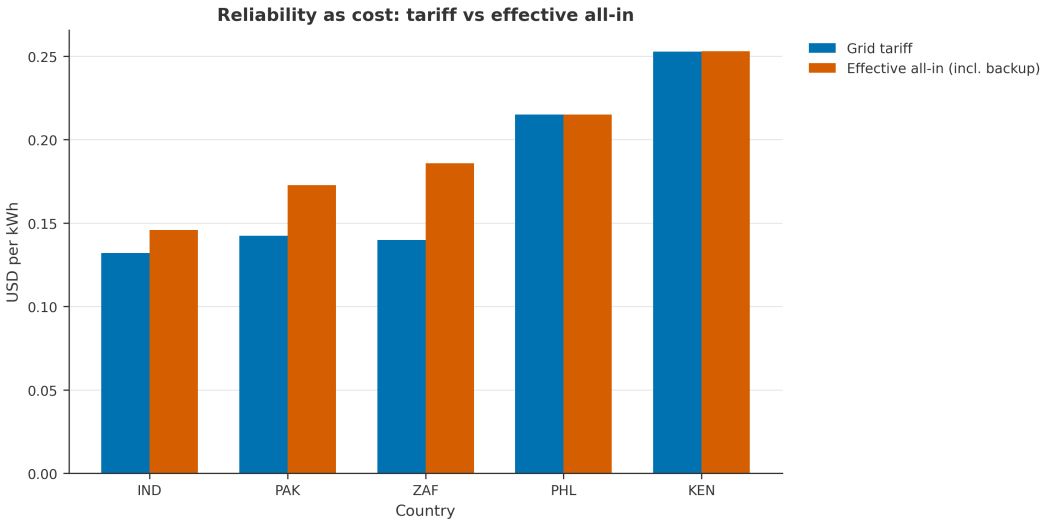


Figure 11. Tariff vs effective all-in cost including backup generation at $0.45/kWh (illustrative). Nigeria excluded: no public national SAIDI; firm evidence implies 5–10x the tariff.

Connection delays are a second reliability cost, and the binding constraint is financing, not interest. Verified from the time-to-power paper (Kilambi Research, 2026a): a 12-month delay on the stylised $6B data centre costs $857.6M (14.3% of capex); delayed revenue is about 40% of the total; interest-only accounting understates delay cost by two-thirds. Getting-electricity times (structural, vintage 2008–2011, series since discontinued): Nigeria 260 days, Pakistan 266, South Africa 214, against Germany 17 and Singapore 36. For a financed warehouse or mall, six to nine months of carry plus shifted lease-up is the mechanism, and the wait-versus-solve breakeven bounds the rational premium for captive generation.

The falsifier for this section was whether backup-adjusted costs differ negligibly from tariffs in the worst-reliability countries. For Nigeria and South Africa the falsifier fails and the reliability framing stands; for Kenya and the Philippines it succeeds and the paper reports the tariff as the burden. Both outcomes are in the results, which is what a falsifier is for.

8 Volatility and predictability

Figure 6 shows industrial price paths for eight Asian economies, 2016–2026 in USD. Sri Lanka's crisis whiplash (+75%, +66%, then -20%) is the sample's sharpest discrete repricing; Pakistan's

USD path is FX-driven; China's is flat by administration; Singapore's rises with fuel pass-through (Observed, B/C). Only eight countries have complete verified 2016/2021/2026 USD industrial series; European histories are mostly grade C and several pre-2026 rows were not separately verified, so the paper does not plot them.

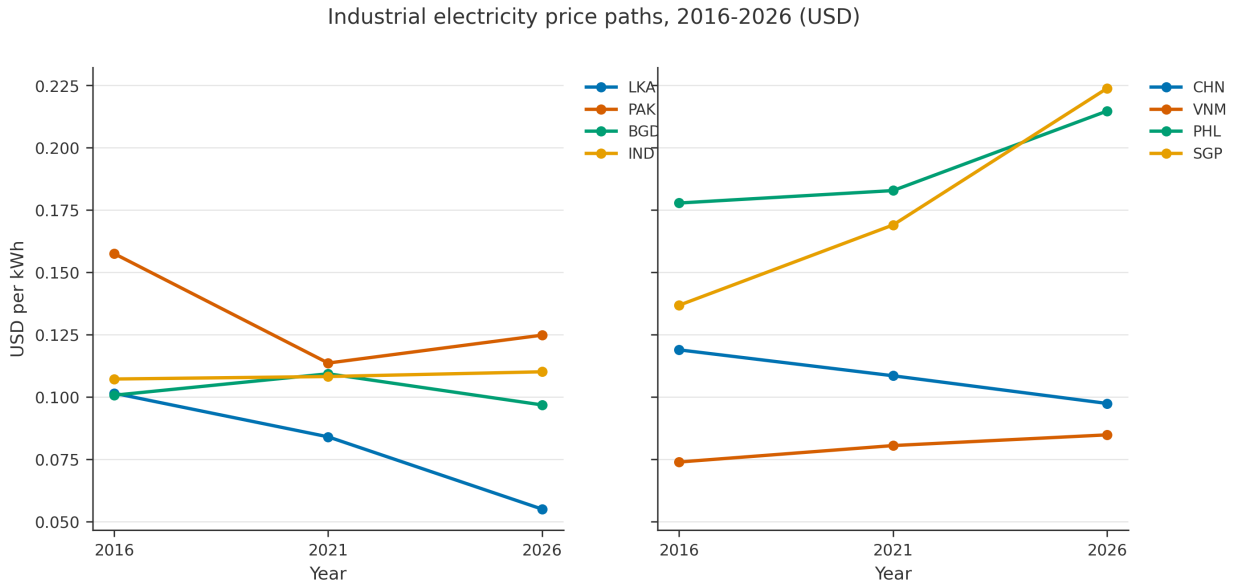


Figure 12. Industrial electricity price paths, 2016–2026, USD/kWh. Shared y-axis. Sri Lanka shows the sample's sharpest crisis repricing; China's path is administered flat.

The 41-market volatility scorecard records how customers actually experience price risk: wholesale exposure (high/medium/low), the dominant bill-volatility driver, demand-charge share where known, time-of-use structure, and corporate PPA market depth. Four findings stand out.

First, in PJM the average price is the shock that a volatility event created. Capacity cleared from $28.92 to $329.17/MW-day (at the FERC cap); data-centre load drove 63% of the 2025/26 $9.3bn increase. Illustrative arithmetic (Calculated, B): about $14.7/MWh added for flat-load industry at 85% load factor, roughly 17% of the US average industrial price. The industrial class allocation share is a missing datum, flagged as such.

Second, Korea's frozen fuel-cost adjustment is the sample's clearest discrete tariff risk: +5 won/kWh held for 18 quarters against a +7.3 won/kWh formula, with the system marginal price up 43.8% over January–August 2026 and KEPCO debt at 210.7tn won. Normalisation is politically sensitive ahead of the 2028 election (Observed, B/C).

Third, the largest forward swing is gas pass-through mechanics, not the gas price itself. The IEA STEPS-vs-CPS spread is ~$2.8/MBtu at Japan by 2035, but Japan's fuel-cost adjustment (four-month lag plus ceilings), Korea's frozen gap, Malaysia's ICPT, and Bangladesh's BPDB losses amplify it into bills (International Energy Agency, 2025b).

Fourth, data-centre burden channels differ by market and must not be averaged: the United States transmits it through capacity and transmission cost recovery onto bills; Ireland through connection policy; Malaysia through RP4 capex in the tariff base; the Nordics through surplus erosion (53 to 29 TWh by 2030) plus Denmark's connection pause and Sweden's queue jumping from 1,310 to 6,700 MW.

9 Forward scenarios: five to ten years

Three scenarios with explicit assumption tables (base, high-cost, low-cost) are set out in the program notes; all are (MODELLED, C) and no point forecast is presented as a prediction. The load-bearing assumptions, each re-verified against producer sources rather than carried over:

- Electricity demand grows ∼40% to 2035 (IEA WEO 2025), with data centres the fastest increment; the absorbability paper's scenarios bound AI-load growth and its $1.79/GPU-hour cost floor (International Energy Agency, 2025b; Kilambi Research, 2026b).
- A 65 bcm LNG overhang by 2030 is the main low-cost tailwind, but it helps Asia more than Europe: coal-to-gas switching needs $5–6/MMBtu delivered, which LNG exporters struggle to supply profitably.
- Carbon policy step-changes: EU ETS tightening plus CBAM, UK ETS, Korea ETS, China ETS expansion, Japan GX. The industries facing step changes are the energy-intensive set of Section 6.1.
- Transmission and grid-investment cost recovery (who pays: rate base vs taxpayers) is the under-modelled wedge in most markets.
- Currency and industrial-policy scenarios for the vulnerable set: Turkey, Argentina, Pakistan, Nigeria, Egypt.

The stated falsifier for the high-cost scenario: if the LNG overhang arrives on schedule and the EC ETS overhaul is adopted, both drivers unwind together and the high-cost case is less robust than it looks. The paper keeps the scenario and reports the fragility.

10 Falsifiers and limitations

The shared falsifiers file (`data/SHARED_FALSIFIERS.md`) states eight advance tests; the verdicts are reported in the sections above. Three limitations deserve emphasis here because they bound what the paper can claim.

First, the burden index is a transparent composite with arbitrary equal weights. Reweighting toward price restores the nominal ranking; reweighting toward reliability promotes Nigeria and South Africa further. The index is a disciplined way to display the wedges, not a discovery of the "true" ranking.

Second, the country is sometimes the wrong unit. Bilateral and captive contracting (Section 6.1), provincial spreads (China 2–3x), and US regional spreads (3.3x) all mean the national tariff misdescribes the marginal plant. The paper flags each case rather than averaging through it.

Third, the dataset's largest holes are named in Section 3: no industrial coal or coke prices, no demand-charge bill shares, five countries without SAIDI, seven economies without industrial all-in prices, and several European histories at grade C. Results that depend on those inputs are graded accordingly and the grades are printed, not buried.

11 Conclusion

The nominal tariff is the beginning of the burden calculation, not the end. Across 54 economies, the wedges between posted price and effective burden, delivered charges, reliability and backup, subsidy positions, intensity and productivity, volatility, are large enough to reorder the rankings that tariff tables produce. The organising hypothesis survives with its limits stated: low-nominal-price economies do face greater effective burden where the grid fails or the subsidy is a fiscal transfer (Nigeria, South Africa, Pakistan, Ecuador), and high-price economies do partially offset cost through efficiency and economic structure (Ireland, Germany, Singapore, Taiwan). The offsets are partial, sector-specific, and sometimes absent (Australia). Systematically, the rank correlation between nominal industrial price and income-adjusted burden is 0.30, and against the composite burden index 0.01: nominal rankings carry almost no information about who bears the heaviest real burden. For aluminium the tariff is the whole story and the national average is the wrong number; for semiconductors the tariff barely matters; for dairy the gas price matters more than the electricity price; for data centres the tariff is the business case and efficiency cannot close the gap. The paper's contribution is the measurement: all-in, reliability-adjusted, subsidy-aware, sector- translated, with the grades and the gaps printed alongside the numbers.

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A Country burden indicators

Table 1 reports the composite burden index and its inputs for the 47 ranked countries. Full machine-readable files (108 snapshot rows, 247 history rows, sector sheets, volatility scorecard) are in the program dataset.

Table 1. Burden index and components, 47 countries. Outage hours: snapshot value with reliability-file or SAIDI fallback; n.a. where no source reports a value. Nigeria's 4,500 hours is a worker estimate (no public national SAIDI) and is treated as indicative.

ISO	Country	Index	Price ($/kWh)	Outage h/yr	Grade
UKR	Ukraine	88.9	0.2015	n.a.	B
NGA	Nigeria	75.0	0.1483	4500.0	B
GEO	Georgia	73.6	0.0945	n.a.	C
ZAF	South Africa	72.5	0.0866	300.0	B
PAK	Pakistan	69.8	0.1250	300.0	Observed B
KAZ	Kazakhstan	68.9	0.0972	n.a.	C
KEN	Kenya	65.7	0.1864	122.0	B
CHL	Chile	65.7	0.1990	24.1	C
AZE	Azerbaijan	65.6	0.0529	n.a.	C
GHA	Ghana	65.4	0.1795	50.0	B
BRA	Brazil	65.2	0.1870	24.1	C
COL	Colombia	64.9	0.2310	24.1	C
IND	India	63.4	0.1101	50.0	Observed B
UZB	Uzbekistan	62.3	0.0711	n.a.	C
BGD	Bangladesh	62.2	0.0968	600.0	Observed B
AUS	Australia	61.3	0.2660	1.9	C
VNM	Vietnam	60.8	0.0700	n.a.	B
ISR	Israel	58.6	0.1738	15.0	C
RUS	Russia	57.3	0.0488	n.a.	C
TUR	Turkiye	56.5	0.1249	20.0	B
ECU	Ecuador	55.9	0.1020	24.1	C
OMN	Oman	55.0	0.0628	10.0	B
PER	Peru	54.9	0.1100	24.1	C
MAR	Morocco	54.2	0.1123	15.0	C
PHL	Philippines	54.1	0.2150	1.8	B
LKA	Sri Lanka	53.9	0.0906	150.0	Observed B
ARE	United Arab Emirates	53.9	0.0943	5.0	C
ARG	Argentina	53.8	0.0980	24.1	C
SAU	Saudi Arabia	51.4	0.0613	10.0	B
IDN	Indonesia	51.0	0.0620	n.a.	B
NZL	New Zealand	49.7	0.0990	4.8	C
USA	United States	49.4	0.0861	11.0	Observed B
CAN	Canada	48.1	0.0620	6.0	Observed C
GBR	United Kingdom	47.9	0.2275	0.7	B
CHN	China	47.5	0.0974	3.0	Observed B
JPN	Japan	46.8	0.1690	0.6	B
MYS	Malaysia	46.2	0.1110	0.8	B
MEX	Mexico	45.7	0.2052	0.3	Calculated B
THA	Thailand	45.6	0.1270	0.4	B
SWE	Sweden	45.3	0.1096	1.7	B
SGP	Singapore	45.0	0.2690	0.0	B
DEU	Germany	43.9	0.2558	0.2	B
KWT	Kuwait	42.8	0.0163	10.0	B
KOR	South Korea	42.6	0.1340	0.2	B
TWN	Taiwan	39.4	0.1180	0.2	B
IRL	Ireland	39.4	0.2884	3.6	B
QAT	Qatar	34.3	0.0192	5.0	C

B Income-adjusted burden indicators

Table 2. Income-adjusted burden indicators, 47 countries. Cost of 1 MWh at the industrial all-in tariff as a share of GDP per capita (nominal and PPP), and economy-wide electricity cost per $1M of manufacturing value added as a proxy share of MVA (see Section 2 for why this is a proxy, not a budget share).

ISO	Country	$/kWh	MWh/GDPpc nom (%)	MWh/GDPpc PPP (%)	Elec/$1M MVA (%)
KEN	Kenya	0.1864	11.99	4.76	25.94
GHA	Ghana	0.1795	9.39	3.58	38.97
PAK	Pakistan	0.1250	8.45	2.00	32.8
NGA	Nigeria	0.1483	7.16	2.10	15.41
PHL	Philippines	0.2150	5.39	1.82	36.88
IND	India	0.1101	4.25	1.03	37.87
BGD	Bangladesh	0.0968	3.73	1.00	10.16
MAR	Morocco	0.1123	3.58	1.43	21.07
UKR	Ukraine	0.2015	3.44	1.07	120.0
COL	Colombia	0.2310	2.70	1.02	46.72
LKA	Sri Lanka	0.0906	1.99	0.58	7.39
UZB	Uzbekistan	0.0711	1.79	0.52	20.89
BRA	Brazil	0.1870	1.75	0.80	53.1
THA	Thailand	0.1270	1.72	0.51	22.96
ZAF	South Africa	0.0866	1.53	0.64	54.19
VNM	Vietnam	0.0700	1.48	0.43	18.55
MEX	Mexico	0.2052	1.48	0.79	19.66
ECU	Ecuador	0.1020	1.43	0.61	21.73
IDN	Indonesia	0.0620	1.26	0.38	8.71
TUR	Turkiye	0.1249	1.14	0.47	23.37
PER	Peru	0.1100	1.14	0.58	18.63
CHL	Chile	0.1990	1.11	0.53	55.04
GEO	Georgia	0.0945	0.97	0.32	47.1
MYS	Malaysia	0.1110	0.94	0.29	23.02
CHN	China	0.0974	0.73	0.36	19.27
AZE	Azerbaijan	0.0529	0.71	0.20	31.23
KAZ	Kazakhstan	0.0972	0.66	0.22	31.51
ARG	Argentina	0.0980	0.66	0.30	17.15
JPN	Japan	0.1690	0.50	0.31	21.82
ISR	Israel	0.1738	0.46	0.46	30.04
DEU	Germany	0.2558	0.42	0.34	14.67
AUS	Australia	0.2660	0.41	0.37	80.7
GBR	United Kingdom	0.2275	0.40	0.35	21.58
KOR	South Korea	0.1340	0.37	0.22	16.8
OMN	Oman	0.0628	0.37	0.19	36.1
TWN	Taiwan	0.1180	0.34	0.15	11.58
SGP	Singapore	0.2690	0.28	0.18	17.83
RUS	Russia	0.0488	0.28	0.10	16.35
SAU	Saudi Arabia	0.0613	0.28	0.12	23.51
ARE	United Arab Emirates	0.0943	0.22	0.14	35.41
IRL	Ireland	0.2884	0.22	0.19	4.06
NZL	New Zealand	0.0990	0.20	0.18	20.87
SWE	Sweden	0.1096	0.17	0.15	17.05
CAN	Canada	0.0620	0.11	0.09	19.45
USA	United States	0.0861	0.10	0.10	15.13
KWT	Kuwait	0.0163	0.06	0.04	14.5
QAT	Qatar	0.0192	0.03	0.02	6.33

C Sector energy intensities

Table 3. Typical energy intensities per unit of physical output, energy-intensive industries. Gas intensity is the gas-route figure; coal-route production (notably China) differs (see Section 6.1).

Product	Typical (MWh/unit)	Range	Gas (MMI
crude steel, BF-BOF route (t crude steel)	0.35	0.15–0.6	
crude steel, EAF route (100direct reduced iron, NG shaft furnace (t DRI)	0.15	0.1–0.25	
primary aluminium, Hall-Heroult smelting (t Al)	14.5	13.0–16.0	
chlorine, membrane chlor-alkali (t Cl2)	2.6	2.1–3.0	
ammonia, NG steam methane reforming (t NH3)	0.3	0.1–0.5	
urea (from ammonia) (t urea)	0.15	0.08–0.25	
clinker (dry process kiln) (t clinker)	0.07	0.05–0.1	
chemical kraft pulp, bleached market (ADt)	0.7	0.55–0.85	
paper and board, average grade mix (t paper)	0.5	0.35–0.7	